

Supporting Evidence

Attachment Rescue · Track 1 (Ideas). All figures reproducible from public data with the linked code.

1. CENSUS OF THE FULL POPULATION (FILES LISTED BY NAME, NOTHING DOWNLOADED)

NCI-funded open-access papers, 2016–2022, with supplementary files and no data-repository accession	Count
Papers in population (Europe PMC, funder "NCI NIH HHS"; no text-mined accession for GEO, SRA, dbGaP, EGA, ArrayExpress, PRIDE, BioProject, MetaboLights, PDB, GenBank)	26,228
Papers whose full text was readable · attachments listed	25,138 · 73,795
Papers with at least one spreadsheet / CSV / tab-text attachment	4,759 (18.9%)
Spreadsheet / CSV / tab-text files (Excel alone 14,448; 15.9 GB)	15,458
Those papers licensed CC BY or CC0 (files may be re-deposited unchanged)	3,824 (80%)
Same census, 300-paper samples from other institutes: NHLBI 14.9% · NIDDK 17.6% · NIMH 18.7% · NIA 20.5% · NINDS 23.8% (NCI by the same yardstick: 17.9%); those five populations total 63,043 papers	NIH-wide

2. FOUR RANDOM SAMPLES OF 300 PAPERS, EVERY ATTACHMENT OPENED (124 WITH A GEO ACCESSION EXCLUDED)

Cancer type	Analysed	Any dataset*	Native spreadsheet	CC BY / CC0	Dataset files
Glioblastoma	271	53 (19.6%)	41 (15.1%)	41	106
Breast cancer	271	51 (18.8%)	41 (15.1%)	46	132
Colorectal cancer	273	53 (19.4%)	41 (15.0%)	47	112
Leukemia	261	45 (17.2%)	33 (12.6%)	33	112
Pooled	1,076	202 (18.8%)	156 (14.5%, CI 12–17%)	167 (83%)	462

*Dataset = largest table ≥ 50 rows × 3 cols, or ≥ 1,000 rows; PDF/Word tables only as numeric grids; review files, reporting summaries and figures excluded. Largest table 152,683 rows. Classifier decisions re-examined on 90 files with the AI assistant, not blind, verdicts published: 60/60 "dataset" calls real; every miss under 50 rows. About half primary measurements, half derived results. Filter check on 500 random papers: 15% reference a repository somewhere; 13.5% of spreadsheet papers already have a Figshare identifier. Net estimate: 2,800–3,900 papers, 8,000–11,000 buried dataset files.

3. TWO LIVE EXAMPLES A REVIEWER CAN OPEN

- **121-patient glioblastoma cohort table** (sex, age, location, IDH/TERT, MGMT, ATRX) — *Acta Neuropathol Commun* 2018, PMC6193287, 40478_2018_613_MOESM2_ESM.xlsx. CC BY.
- **118,273-row drug-screening dataset** — *Nat Commun* 2019, PMC6599391, 41467_2019_10912_MOESM4_ESM.xlsx. CC BY.

4. RESCUED TABLES PUT BACK TO WORK

- Two rescued TCGA-derived glioma tables joined by patient ID: 244 shared patients agree on survival, IDH and age in >96%; pooled 1,037 patients reproduce the known IDH survival split (median 88 vs 15 months; validation only). A third rescued table is a 121-patient hospital cohort not found in any catalogue (demo/).
- Verified patient-level tables in the four samples: ~4,400 patient records, ~500 from institutional cohorts not found in TCGA, GEO, dbGaP or any catalogue. 8 datasets packaged with DataCite metadata, not yet uploaded (Prompt 4).

5. CODE, DATA, SITE (OPEN, MIT)

- Repository: github.com/OJ-IRO/attachment-rescue — code, raw census, per-cancer results, hand-check verdicts, packages.
- Public page with searchable list of the 462 dataset files and plain-language descriptions: attachment-rescue.vercel.app

6. PRIOR WORK CITED

- Gobeill et al., *Bioinf Adv* 2025, 36.2 M PMC supplementary files indexed (PMC12371329).
- Federer 2018 (pone.0194768); Colavizza 2020 (pone.0230416); Federer 2022 (PMC9401135).
- Ghosh 2025 (PMC12782234); Boja 2026 (PMC13479720); Boyd 2026 (jnci/djag199).
- CBIIT: github.com/CBIIT/INS-Data, ccdi-ins-icdo-mapping-llm.